SIMATS ENGINEERING

ASSESSMENT TOOL 2: Industry Problem Solving Task

COURSE NAME:	Cloud Computing and Big Data Analytics	**COURSE CODE:**	CSA15

COURSE OUTCOME COVERED

CO4: *Analyze big data characteristics and challenges across domains including security and application aspects.* (BL4)

Dave's Taxonomy: Articulation (Level 4) **Total Marks: 40**

Question Count: 5

Duration :60 Minutes

Code of Conduct

I Pragadeesh Nalan S V (192321161) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: Pragadeesh Nalan S V

Assessment Tasks

Industry Problem Solving Task

1. **Retail Data Optimization Problem -** A retail chain operates both physical stores and an online platform. It generates large volumes of sales, inventory, and customer behavior data. The company wants real-time inventory tracking and demand forecasting. It also aims to deliver personalized marketing recommendations to customers. Design a big data architecture to support these requirements. Explain the tools, technologies, and analytics models you would use.

2. **Social Media Fraud Detection Problem -** A social media platform processes billions of user interactions daily. These include posts, comments, likes, and shares. The platform wants to detect fake accounts and misinformation quickly. Real-time or near real-time processing is required. Design a scalable big data solution for this use case. Explain the role of machine learning and stream processing frameworks.

3. **Government Data Platform Problem** A government agency is building a national data platform. It combines census, economic, and geospatial datasets. Different departments need access to shared and integrated data. The system must ensure interoperability across multiple agencies. Strict privacy and security requirements must be followed. Propose a governance and data management strategy.

4. **Banking Fraud Detection Problem**
   A bank processes millions of financial transactions every minute. It needs to detect fraudulent activities in real time. The system must analyze historical and streaming transaction data. Low latency and high accuracy are critical requirements. Design a big data pipeline for fraud detection. Explain the technologies and algorithms you would use.

5. **Agriculture Smart Farming Problem**
   A smart farming system collects soil, weather, and crop data. Sensors and drones provide real-time field information. Farmers want insights to improve yield and resource usage. Data is generated from multiple distributed sources. Design a big data solution for precision agriculture. Explain how analytics can support decision-making.

Industry Problem Solving Rubric

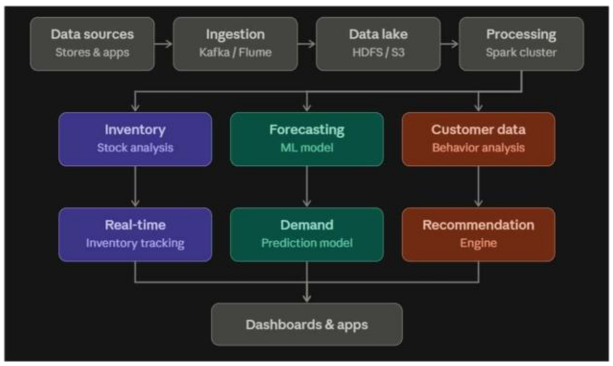
Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Understanding of Industry Context	20%	Demonstrates strong understanding of real-world problem context	Good understanding	Basic understanding	Weak understanding
Application of Big Data Concepts	25%	Applies highly relevant big data ideas and tools	Mostly appropriate application	Partial application	Weak application
Solution Design	25%	Solution is practical, scalable, and well-reasoned	Reasonable solution	Basic solution with gaps	Unclear solution
Security / Risk Awareness	15%	Strong awareness of security and operational risks	Reasonable awareness	Limited awareness	Poor awareness
Clarity and Professionalism	15%	Well-structured and professional response	Generally clear	Moderately clear	Poorly presented

Total Marks Awarded:

Industry Problem Solving Task

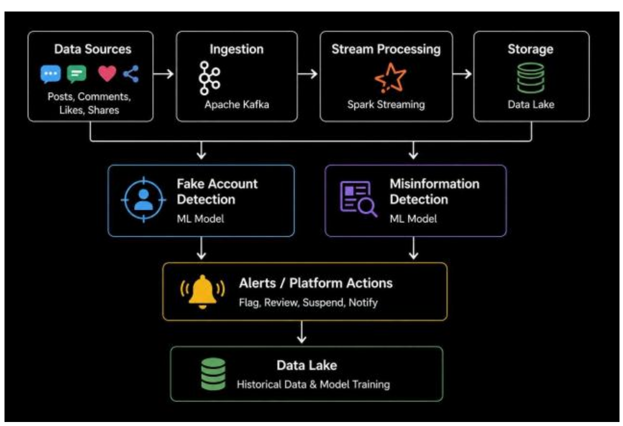
1. Retail Data Optimization Problem

A big data architecture can use **Kafka** to collect real-time sales, inventory, and customer data from stores and online platforms. **HDFS/Cloud Data Lake** can store large volumes of historical data, while **Apache Spark** can process both batch and streaming data. **NoSQL databases** can provide fast access to inventory and customer information. Machine learning models such as **time-series forecasting** can predict product demand, while **clustering and recommendation algorithms** can provide personalized offers. Dashboards using **Power BI/Tableau** can monitor inventory and sales in real time. This helps reduce stock-outs, improve inventory management, and increase customer engagement.



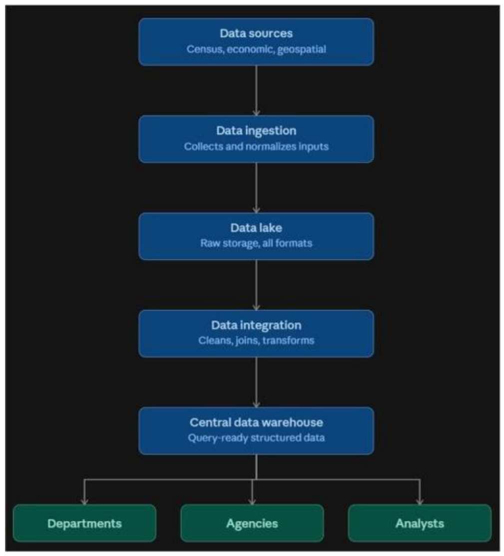
2. Social Media Fraud Detection Problem

The platform can use **Apache Kafka** to collect posts, comments, likes, and shares as real-time streams. **Spark Streaming** can process these interactions and extract useful features. Historical data can be stored in a **Data Lake/HDFS**. Machine learning models such as **Random Forest, Logistic Regression, and Gradient Boosting** can identify fake accounts using account age, posting frequency, followers, and interaction patterns. **NLP techniques** can analyze text to identify potentially misleading or false information. Suspicious accounts and content can then be flagged automatically, allowing rapid action.



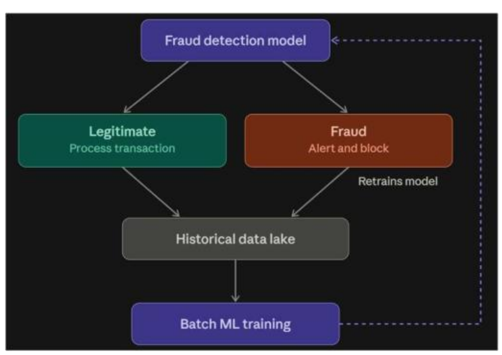
3. Government Data Platform Problem

The government can build a centralized **Data Lake** to combine census, economic, and geospatial datasets. **ETL/ELT tools** can clean, transform, and integrate data from different departments. A **data catalog and common data standards** can improve interoperability and make datasets easier to discover. **Role-Based Access Control (RBAC)** can restrict access according to user roles. Sensitive information should be encrypted and anonymized, with audit logs maintained for security. Data governance should define ownership, quality standards, metadata, data lineage, and retention policies. This provides secure and consistent data sharing between government agencies.



4. Banking Fraud Detection Problem

A banking fraud system can use **Kafka** to collect millions of transactions as real-time streams. **Spark Streaming** can process transactions with low latency, while historical transactions are stored in a **Data Lake** for model training. Machine learning algorithms such as **Logistic Regression, Random Forest, Gradient Boosting, and Isolation Forest** can identify fraudulent or unusual transactions. Features such as transaction amount, location, time, device, and transaction frequency can be analyzed. Suspicious transactions can immediately generate alerts or be blocked. Continuous model training using historical data improves fraud detection accuracy.



5. Agriculture Smart Farming Problem

A precision agriculture system can collect soil, weather, and crop information using **IoT sensors, drones, and satellites**. **Kafka/MQTT** can transmit sensor data, while a **Cloud Data Lake** stores historical information. **Apache Spark** can process large-scale sensor and image data. Machine learning can be used for **crop-yield prediction, disease detection, irrigation planning, and field classification**. Dashboards can provide farmers with real-time information about soil moisture, weather, and crop health. This helps farmers optimize water and fertilizer usage, detect diseases early, and improve crop yield.